# 4.2.3 Configuring a style

By default, all features of a vector layer get the same style (A, *Single Symbol*): all points get the same marker, all lines the same line style, all polygons the same fill. You can configure in detail what this style looks like. As you saw on the previous page, a symbol consists of one or more overlapping symbol layers. For each symbol layer you choose a **type** (*Symbol layer type*). Below we discuss three widely used kinds in more detail:

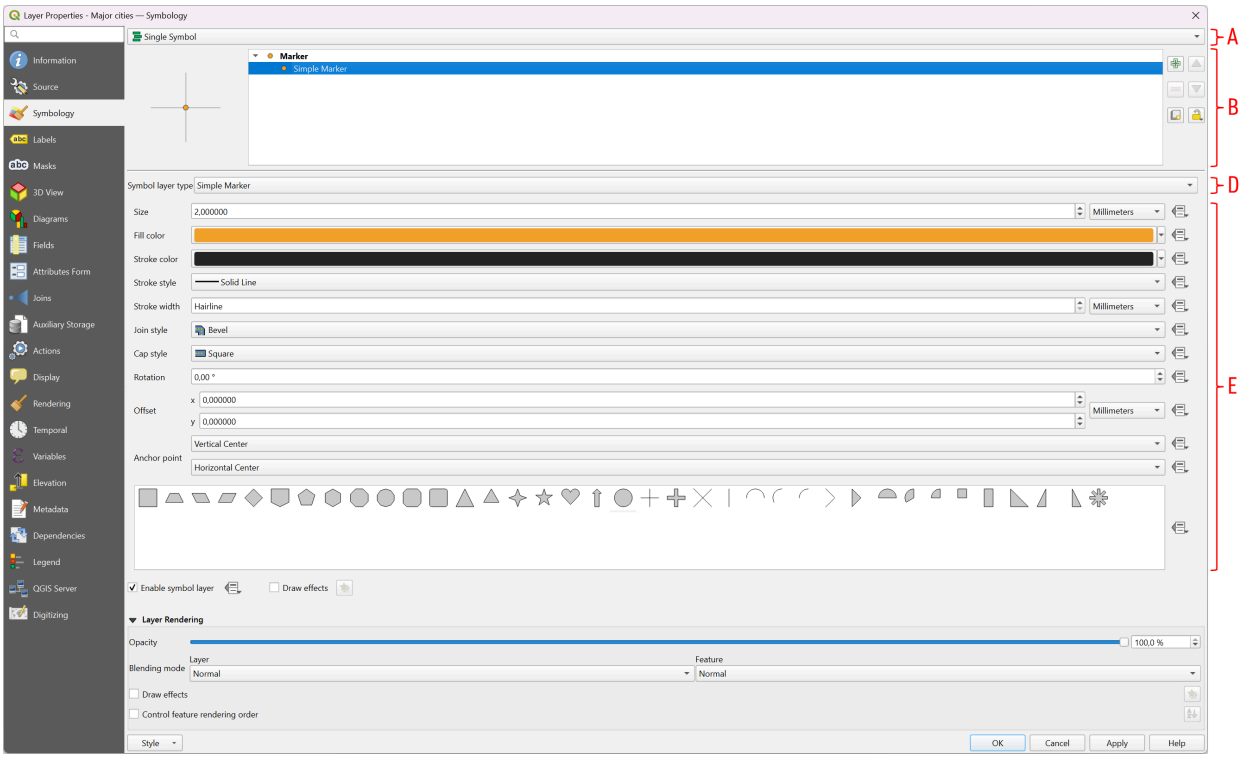
- **Simple symbols**: basic shapes that come with QGIS. These are the default types, and also the ones you will use most often.
  - For points, *Simple Marker*: a geometric shape such as a circle, square, triangle or cross, of which you set the colour, size, outline…
  - For lines, *Simple Line*: a line of which you set the colour, width, dash pattern…
  - For polygons, *Simple Fill*: a fill of which you set the colour, hatching, opacity…, with an outline for which you can set the same wide range of properties as for a line (colour, width, dash pattern…).
- **Image symbols**: instead of a geometric shape, you use an image, as a point symbol (*SVG Marker*, *Raster Image Marker*) or as the fill of a polygon (*Raster Image Fill*). You can choose an image from the library that comes with QGIS (for SVG symbols) or import your own.
- **Marker symbols**: for example a marker repeated along a line (*Marker Line*), or a polygon filled with markers (*Point Pattern Fill*).

## Simple symbols

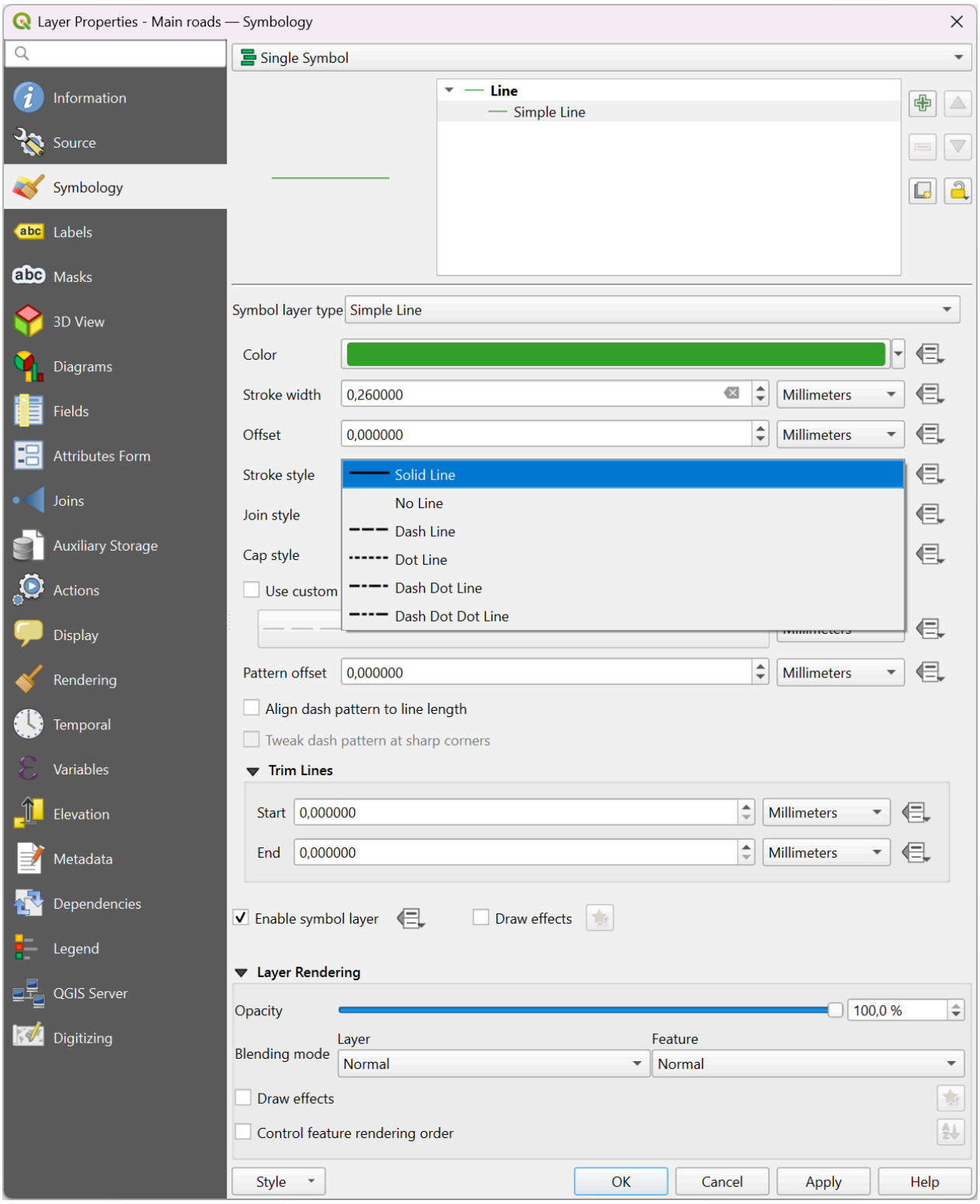
By selecting a symbol layer in the symbol overview (B in the screenshot below), you get access to its configuration options. The first option (D) sets the type of the symbol layer (*Symbol layer type*), and the other options at E depend on that type.

We first explore the options for the simple symbols: *Simple Marker*, *Simple Line* and *Simple Fill*, for points, lines and polygons respectively. These options differ somewhat between the three geometry types, but the principle is always the same. The best way to get to know them is to try them out. The screenshots below give an overview of the configuration options for each type.

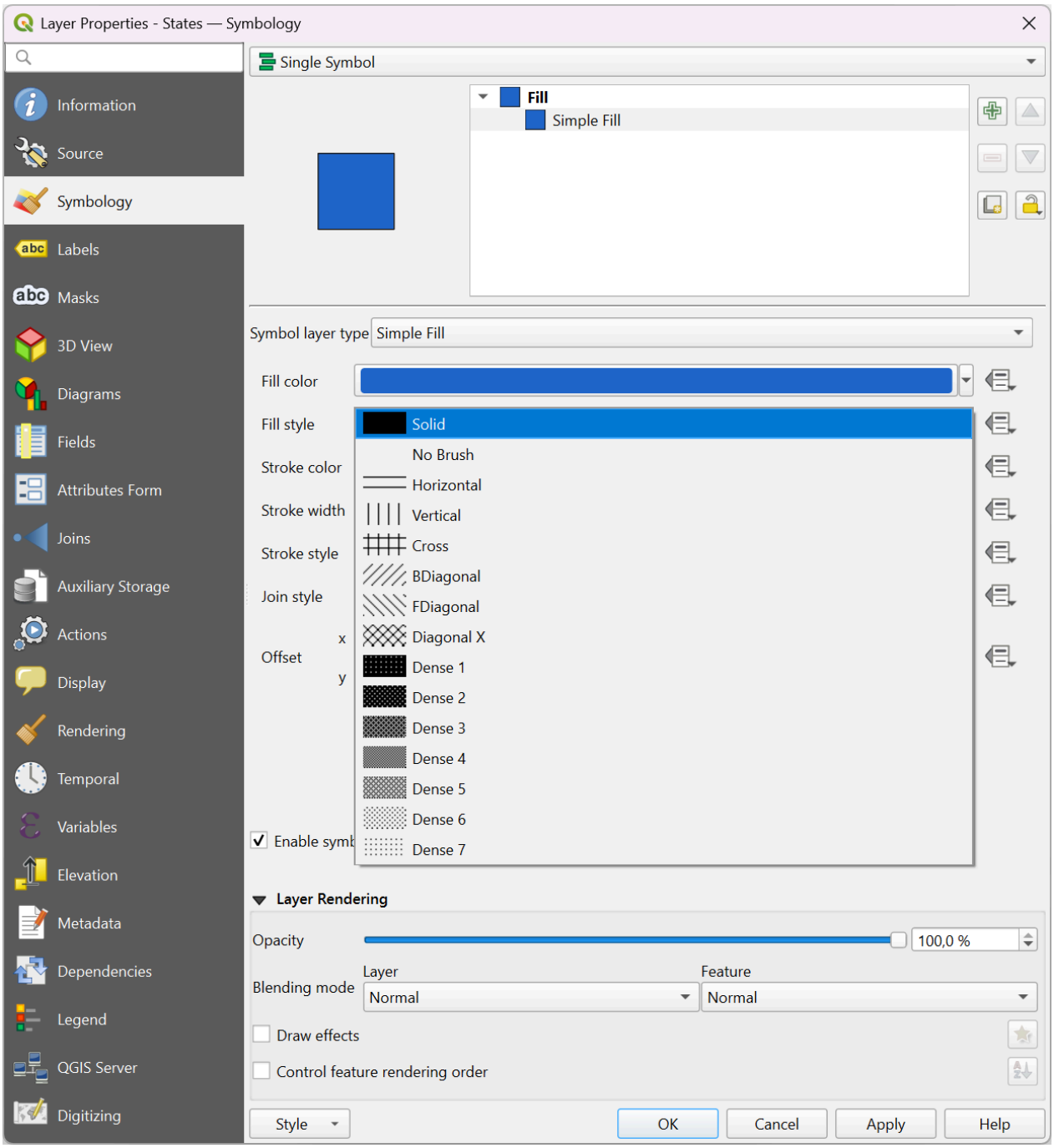
For a **point layer**, one of the most eye-catching configuration options is the shape of the icon used to represent a point.




For **line features**, besides the colour and line width, the dash pattern is often changed.



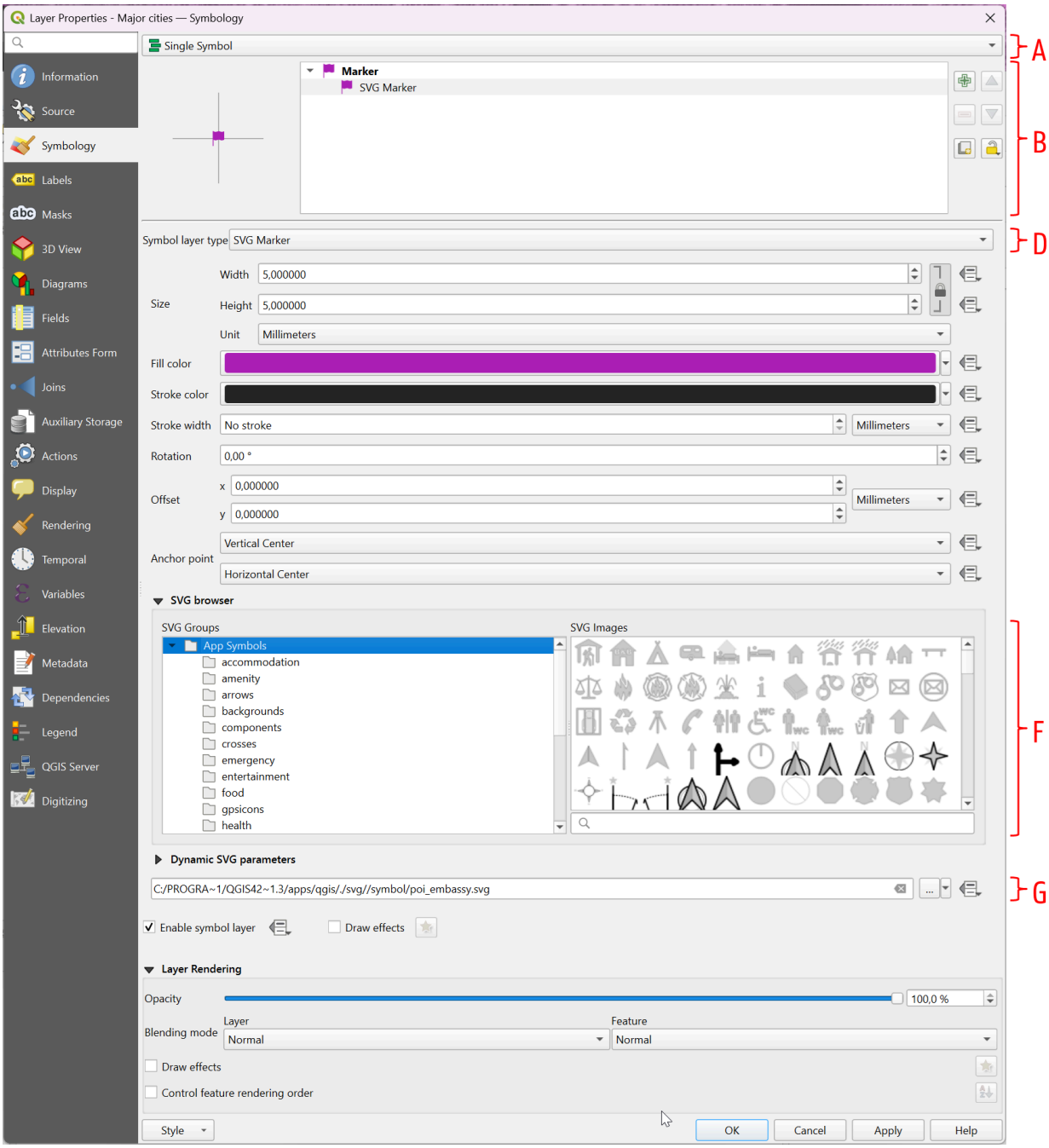

For **polygons**, you can change both the colour and thickness of the borders and the colour and fill pattern of the area. You can also show only the borders of your polygons and leave the area transparent. This can be useful, for example, if you want to visualise several overlapping polygons together in your map canvas.



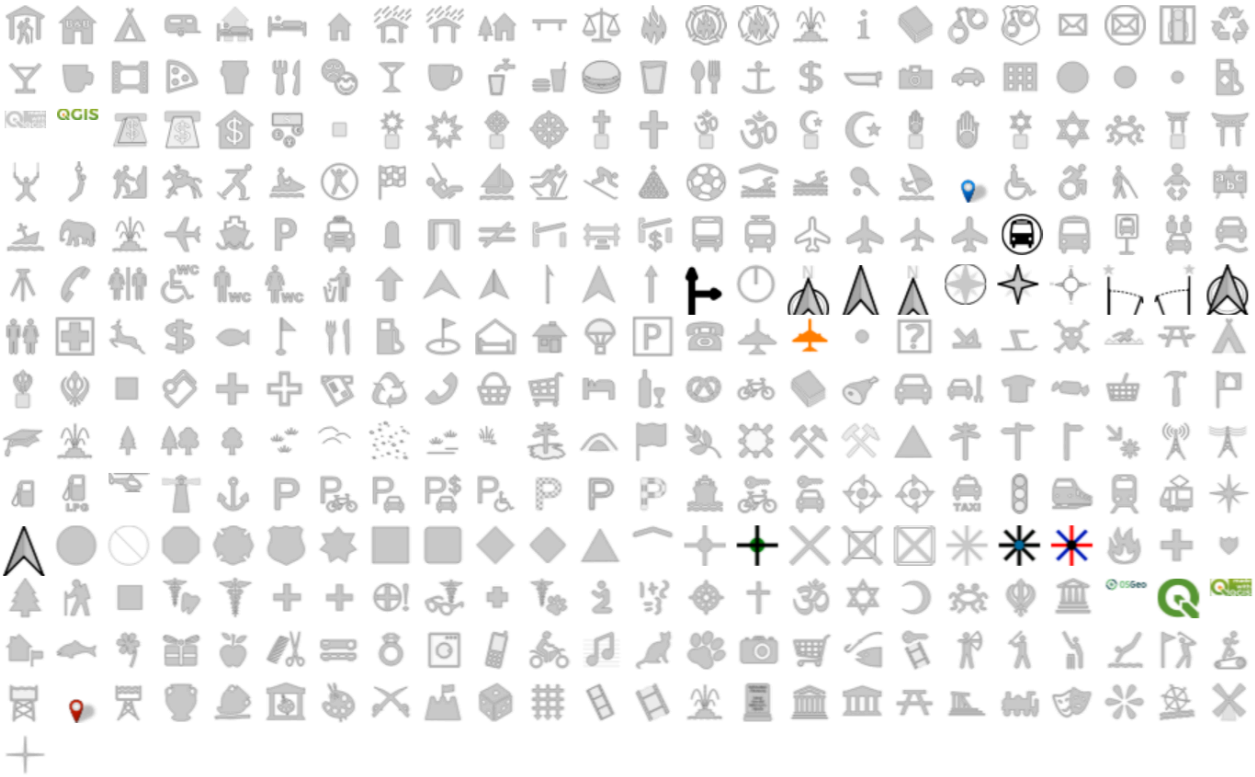

## Image symbols

Sometimes a simple geometric shape is not enough: you want a church, a castle or a battlefield on your map, or a polygon filled with a texture. In that case you use an image instead. For this, QGIS offers, among others, the symbol layer types **SVG Marker** (for images in SVG format) and **Raster Image Marker** or **Raster Image Fill** (for images in raster formats such as PNG or JPG). You choose them in the drop-down list *Symbol layer type* (D).

SVG stands for 'scalable vector graphics'. It is a widely used file format for vector images, such as icons and logos, also outside GIS.




By default, QGIS offers a large variety of typical map symbols in SVG format. You can browse them by theme in the configuration window (see F). Once you have selected the icon you want, you can also change its colour and size, just as with the *simple markers*. The image below shows a selection of the SVG symbols that come with QGIS.

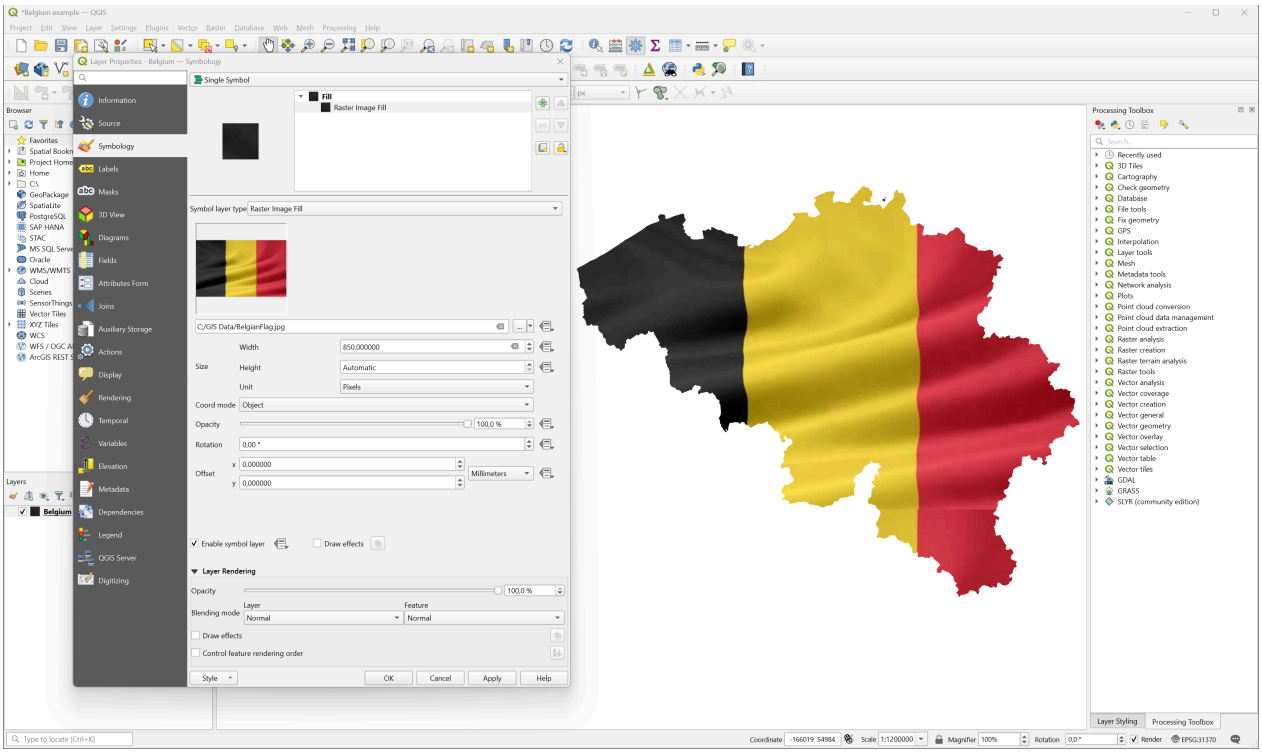


However, you do not have to limit yourself to the symbols that QGIS offers by default. You can also **load your own SVG image (at G)**. A huge number of SVG icons are available on the internet, but make sure you **check the licence** before you download and use them.

If you choose the option *Raster Image Marker* or *Raster Image Fill* under *Symbol layer type*, you can use images in formats such as PNG, TIF or JPG in your symbology. In this way you can choose an image file on your computer and use it as a point icon or as a fill for an area. For example, you could show a layer that represents the territory of Belgium with an image of the Belgian flag.

Warning

**QGIS does not copy the image into your project; it only remembers the path to the image file.** If you move, rename or delete the image, the symbol no longer works (just as with data sources, see The QGIS project). To avoid this, you can store the image in the project itself: click the small arrow next to the file path and choose *Embed File...*.



This *raster image* option is available for point and polygon layers. For line layers there is a similar symbol layer type, *Raster Line*.

## Marker symbols

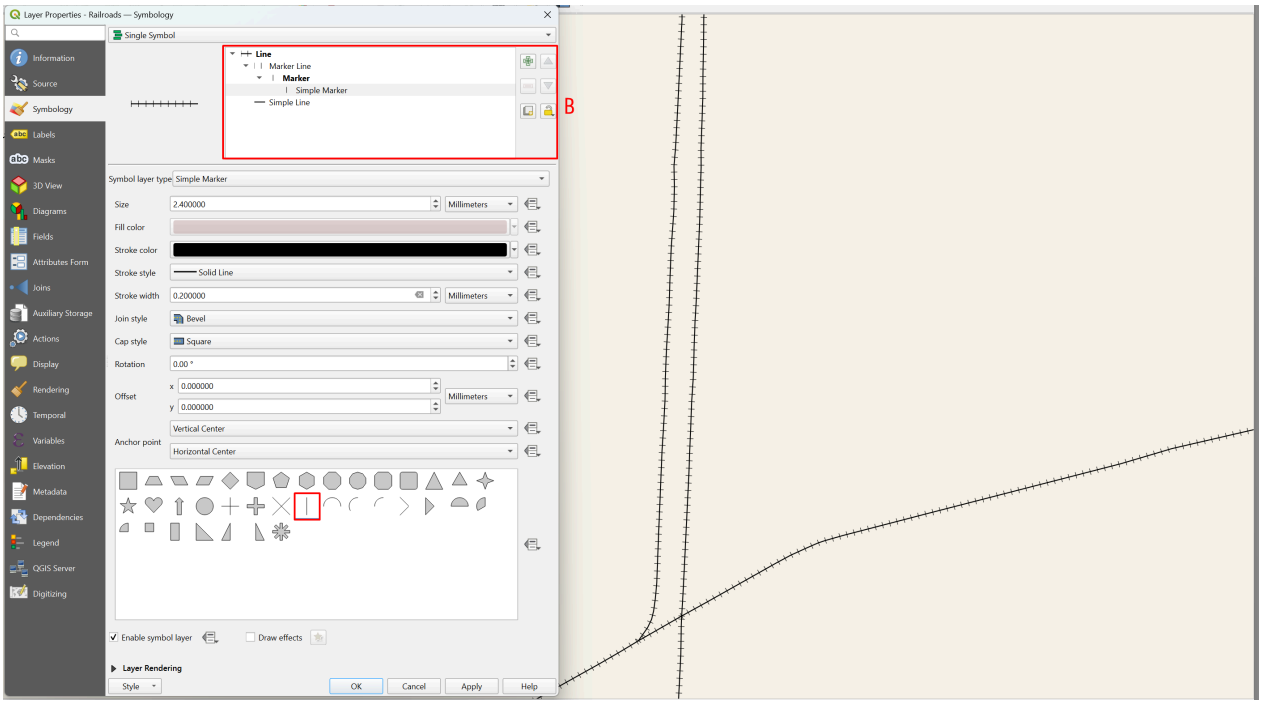
A **marker** is a point symbol. A *Marker Line* repeats a marker along a line, and a *Point Pattern Fill* fills a polygon with markers. You style the marker itself just as you style a point: it can be a *Simple Marker*, but also an *SVG Marker* or any other marker type.

In the symbol overview (B) the marker appears **nested** under the *Marker Line* or *Point Pattern Fill*. You therefore configure two levels:

- If you select the *Marker Line* itself, you set where the markers appear along the line, for example at a fixed interval, on every vertex or only at the start or end of the line.
- If you select the marker nested under it, you set what the markers themselves look like: their shape, size, colour and rotation.

A good example is the default line style *topo railway*, which shows a railway line as on a topographic map. It combines two symbol layers:

- a *Simple Line*: the black line itself;
- a *Marker Line*: its nested marker is a *Simple Marker* with a short vertical stroke as its shape, and the settings of the *Marker Line* place this stroke every few millimetres along the line. Together these strokes form the cross ties of the railway.


Layer Properties - Railroads — Symbology
Single Symbol
Information
Source
Symbology
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3D View
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Dependencies
Legend
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Digitizing
Line
Marker Line
Marker
Simple Marker
Simple Line
B
Symbol layer type
Simple Marker
Size
2.400000
Millimeters
Fill color
Stroke color
Stroke style
Solid Line
Stroke width
0.200000
Millimeters
Join style
Bevel
Cap style
Square
Rotation
0.00 °
Offset
x
0.000000
y
0.000000
Millimeters
Anchor point
Vertical Center
Horizontal Center
Enable symbol layer
Draw effects
Layer Rendering
Style
OK
Cancel
Apply
Help